SAR - Optical Feature Fusion for Agricultural Crop Classification

Project Title: Pixel Forge
Naisha (AU2440181), Ahladita (AU2440243), Priyanshi (AU22440092), Margi (AU2440035)
Department of Computer Science and Engineering
Ahmedabad University

***Abstract*—Agricultural crop classification is important for crop monitoring and agricultural planning. Satellite imagery helps us study crop conditions over large areas. Although, optical satellite images can be disrupted due to clouds, Synthetic Aperture Radar (SAR) provides additional information on the structure of vegetation and surface properties and can also be used in cloudy conditions.**

In this project, we will study Sentinel-1 SAR features and Sentinel-2 optical features using the YieldSAT dataset for Argentina and Brazil. The question we will be studying is if we can get useful information from this combination that differentiates corn from soybean and wheat.

***Index Terms*—SAR, optical data, Sentinel-1, Sentinel-2, NDVI, crop classification, feature fusion**

I. Introduction

Agricultural crop classification is important for crop monitoring and agricultural planning. Satellite imagery helps us study crop conditions over large areas. Although optical satellite images can be disrupted due to clouds, Synthetic Aperture Radar (SAR) provides additional information on the structure of vegetation and surface properties and can also be used in cloudy conditions.

YieldSAT is a dataset for high-resolution crop yield prediction that includes satellite and field data, yield measurements, and environmental data at field level. YieldSAT provides multitemporal Sentinel-2 L2A optical data and crop and field information, yield maps, weather, soil, and topography. Sentinel-1 is not included in YieldSAT, so SAR must be acquired separately.

Chen and Bruzzone studied self-supervised SAR-optical data fusion using Sentinel-1 and Sentinel-2 images. Eisfelder et al. explored cropland and crop-type classification with Sentinel-1 and Sentinel-2 time series using Google Earth Engine for agricultural monitoring in Ethiopia.

The main aims of this project are:

- Analyze the optical data available and derive vegetation parameters like NDVI.
- To get appropriate SAR features such as VV and VH backscatter values.
- To merge optical and SAR data at the same sample points and crop information.
- To compare crop classification using optical only, SAR only, and feature fusion.

II. Methodology

The project is based on a feature-based approach, where optical and SAR data were processed separately and then were fused for crop classification. First, the data is pre-processed, such as cropping area labels, sample records, and data quality. Optical features are then derived from Sentinel-2 data and SAR backscatter features from Sentinel-1 data are explored. These features can then be combined after the sample locations and identifiers have been verified.

A. Optical Feature Extraction

We used the red band (B04) and near-infrared band (B08) to calculate the Normalized Difference Vegetation Index (NDVI). NDVI is a measure of the greenness of vegetation, derived from the difference between near-infrared and red reflectance. It is calculated as:

$$\text{NDVI} = \frac{B08 - B04}{B08 + B04} \tag{1}$$

We computed NDVI for optical data that was available and stored it for future use. The distribution of NDVI values was also analysed using different visualizations in Python in order to get an idea of how NDVI changes across samples.

B. Proposed SAR-Optical Fusion

The backscatter data of the SAR component was accessed through Google Earth Engine and checked with Sentinel-1 VV and VH data. VV represents vertical transmit and vertical receive, while VH represents vertical transmit and horizontal receive. Our next step is to extract these features for the same valid sample locations used in the optical analysis. After verifying the spatial coordinates and sample identifiers, we will combine the optical and SAR features into one feature table.

The resulting feature sets will be compared using optical-only, SAR-only, and fused-feature approaches for crop classification.

III. Results

The optical data from Argentina and Brazil were used to know what data is available and to work out some vegetation features. The total number of records was 100, with 34 corn, 34 soybean, and 32 wheat records. We also checked the records and crop labels, missing values, and duplicate rows.

We calculated the Normalized Difference Vegetation Index (NDVI) using the red band (B04) and near-infrared band (B08). We calculated the NDVI and stored it in `Optical_Features.csv`, and plotted graphs to explore the data. The range of the NDVI values was approximately 0.188 to 0.676, with a mean of 0.357 and a standard deviation of 0.148.

Using the main result of the analysis, a histogram was created illustrating the distribution of NDVI values and a box plot to compare NDVI values between corn, soybean, and wheat. Additionally, the values of optical bands were plotted to see how they change throughout the data. More analysis is required to decide whether NDVI is able to reliably distinguish between the three crop categories.

The structure of the available Sentinel-1 SAR data was also considered. The test raster contained VV and VH bands at a spatial resolution of 10 metres. The average value of VV was approximately -13.91 dB and the average value of VH was approximately -20.08 dB. All of these values are based only on the test raster available and have not yet been validated as measurements from the actual fields in our dataset.

IV. Discussion

The file `SAR_Field_VV_VH_Samples.csv` has 100 rows but no data in VV or VH. The geometry needs examination, some date and year discrepancies exist, and the raster CRS and location need verifying.

The field-level SAR extraction currently returns missing values because the sample geometries need to be further verified. The first feature in each of the 50 Brazil and 50 Argentina records had no geometry. Field-level extraction returned 100 records with no valid VV or VH values.

Before fusion, spatial registration is needed even if field identifiers and temporal registration by acquisition date are available. Different resolutions, dates, and coordinate reference systems complicate alignment. YieldSAT is primarily designed for crop yield prediction, not classification, so labels must be clearly defined and yield should not be used as a crop-class label.

V. Conclusion

Initial analysis of the optical data is done and NDVI is computed to investigate the changes in NDVI values of the samples available. We have also checked the retrieval of Sentinel-1 VV and VH data. The field-level SAR extraction for them, however, currently returns missing values because the sample geometries need to be further verified.

The next steps are to validate the sample coordinates, verify the extracted SAR data, and match the SAR and optical features by using the matching sample locations and sample identifiers. After aligning the data, we will try feature fusion and optical-, SAR-, and fused-feature-based approaches for crop classification.

References

[1] Y. Chen and L. Bruzzone, "Self-Supervised SAR-Optical Data Fusion of Sentinel-1/-2 Images," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, 2022. [Online]. Available: https://iris.unitn.it/handle/11572/372812

[2] C. Eisfelder, B. Boemke, U. Gessner, P. Sogno, G. Alemu, R. Hailu, C. Mesmer, and J. Huth, "Cropland and Crop Type Classification with Sentinel-1 and Sentinel-2 Time Series Using Google Earth Engine for Agricultural Monitoring in Ethiopia," *Remote Sensing*, vol. 16, no. 5, 2024. [Online]. Available: https://www.mdpi.com/2072-4292/16/5/866

[3] L. Liu et al., "Enhanced Crop Classification Through Integrated Optical and SAR Data: A Deep Learning Approach for Multi-Source Image Fusion," 2024. [Online]. Available: https://eprints.gla.ac.uk/301477/

[4] M. Tufail et al., "A Machine Learning Approach for Accurate Crop Type Mapping Using Combined SAR and Optical Time Series Data," 2022. [Online]. Available: https://www.researchgate.net/publication/355158840_A_machine_learning_approach_for_accurate_crop_type_mapping_using_combined_SAR_and_optical_time_series_data

[5] YieldSAT, "A Multimodal Benchmark Dataset for High-Resolution Crop Yield Prediction." [Online]. Available: https://yieldsat.github.io/